

$$\frac{d}{dt} \left( \frac{1}{2} m v^2 \right) = \frac{1}{2} m \frac{d}{dt} (v^2) = \frac{1}{2} m \frac{d}{dt} (v_x^2 + v_y^2 + v_z^2) = \frac{1}{2} m \left( 2 v_x \frac{d v_x}{dt} + 2 v_y \frac{d v_y}{dt} + 2 v_z \frac{d v_z}{dt} \right) = m \left( v_x \frac{d v_x}{dt} + v_y \frac{d v_y}{dt} + v_z \frac{d v_z}{dt} \right) = m \left( v_x a_x + v_y a_y + v_z a_z \right) = m \mathbf{v} \cdot \mathbf{a} = m \mathbf{v} \cdot \frac{d \mathbf{v}}{dt} = m \frac{d}{dt} \left( \frac{1}{2} \mathbf{v} \cdot \mathbf{v} \right) = m \frac{d}{dt} \left( \frac{1}{2} v^2 \right) = \frac{d}{dt} \left( \frac{1}{2} m v^2 \right)$$